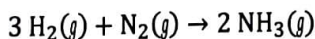


Unit 5 Multiple Choice Practice



1. Gaseous hydrogen and nitrogen combine in the reaction above to form ammonia, NH_3 . Using the table of entropies provided below, calculate the standard entropy change, $\Delta S^\circ_{\text{rxn}}$, for the reaction shown at 25°C .

Substance	S° at 25°C (J/mol K)
$\text{H}_2(\text{g})$	131
$\text{N}_2(\text{g})$	192
$\text{NH}_3(\text{g})$	193

a. -199 J/mol K

b. -386 J/mol K

c. +199 J/mol K

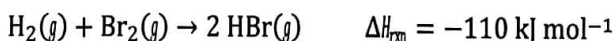
d. +386 J/mol K

$$\begin{aligned} \Delta S &= \sum S^\circ(\text{P}) - \sum S^\circ(\text{R}) \\ &= (3 \cdot 131 + 192) - (2 \cdot 193) \\ &= (2 \cdot 193) - (3 \cdot 131 + 192) = -199 \end{aligned}$$

2. Which of the following is true for an endothermic reaction?

- a. The activation energy is always greater than the activation energy for an exothermic reaction.
 b. Energy is released over the course of the reaction.
 c. The strength of the bonds in the reactants exceeds the strength of the bonds in the products.
 d. The products have more thermal energy than the reactants.

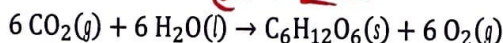
3. If the standard enthalpy of formation for $\text{HBr}(\text{g})$ is -35 kJ mol^{-1} , what is ΔH°_f for $\text{Br}_2(\text{g})$?

a. 0 kJ mol^{-1} b. $-40. \text{ kJ mol}^{-1}$ c. $+40. \text{ kJ mol}^{-1}$ d. $+75 \text{ kJ mol}^{-1}$

$$-110 = (2 \cdot -35) - (\Delta H(\text{Br}_2))$$

tricky!
 Br_2 is (l) @ standard conditions
 so $\Delta H_f \neq 0!$

4. Consider the following reaction showing photosynthesis:



$$\Delta H^\circ = +2800 \text{ kJ/mol}$$

How much thermal energy is absorbed or released when 2.0 moles of glucose, $\text{C}_6\text{H}_{12}\text{O}_6(\text{s})$, is metabolized in the presence of excess oxygen gas, releasing CO_2 and H_2O ?

a. 2800 kJ absorbed

b. 5600 kJ absorbed

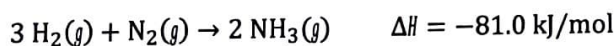
c. 2800 released

d. 5600 kJ released

$$\frac{2}{1 \text{ mol}} \times \frac{2800 \text{ kJ}}{1 \text{ mol}} \times \frac{1 \text{ C}_6\text{H}_{12}\text{O}_6}{1 \text{ rxn}} =$$

(reverse rxn!)

Use the following information to answer the next three questions.



5. Gaseous hydrogen and nitrogen combine in the reaction above to form ammonia, NH_3 . What is the value of the heat of formation of $\text{NH}_3(\text{g})$?

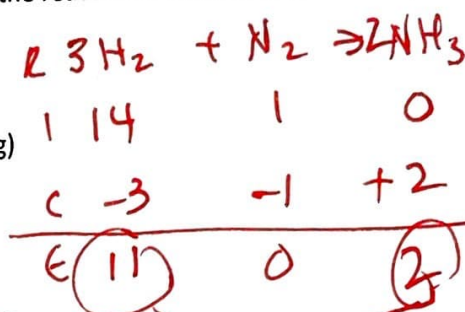
a. -40.5 kJ/mol b. -81.0 kJ/mol c. $+40.5 \text{ kJ/mol}$ d. $+81.0 \text{ kJ/mol}$

$$\Delta H = \sum \Delta H_f(\text{P}) - \sum \Delta H_f(\text{R})$$

$$-81.0 = (2 \cdot \text{NH}_3) - (3 \cdot 0 + 0)$$

6. If 28.0 g of $\text{N}_2(\text{g})$ is combined with 28.0 g of $\text{H}_2(\text{g})$ in a sealed flask and the reaction is allowed to go to completion, what chemical species will still be present inside of the flask?

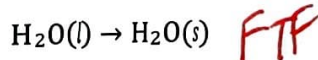
a. $\text{NH}_3(\text{g})$ only
b. $\text{NH}_3(\text{g})$ and $\text{N}_2(\text{g})$
c. $\text{NH}_3(\text{g})$ and $\text{H}_2(\text{g})$
d. $\text{NH}_3(\text{g})$, $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$



7. If 28.0 g of $\text{N}_2(\text{g})$ is combined with 28.0 g of $\text{H}_2(\text{g})$ in a sealed flask and the reaction is allowed to go to completion, how much heat is absorbed or released?

a. 81.0 kJ absorbed b. 375 kJ absorbed c. 81.0 kJ released d. 375 kJ released

$$\frac{1 \text{ mol N}_2 \mid -81 \text{ kJ}}{1 \text{ mol}} = -81 \text{ kJ}$$

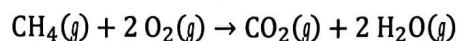


8. Is the process above endothermic or exothermic? Why?

a. Exothermic, because forming new intermolecular attractions between water molecules releases energy.
b. Endothermic, because energy is needed to form new intermolecular attractions between water molecules.
c. Exothermic, because forming new bonds between water molecules releases energy. $\rightarrow$ no new bonds!
d. Endothermic, because energy is needed to form new bonds between water molecules.

No new substance formed!

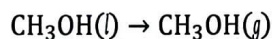
Bond	Average Bond Dissociation Energy (kJ/mol)
C - H	415
O = O	495
C = O	798
O - H	463



9. Use the information provided above to calculate the molar enthalpy of combustion, $\Delta H^\circ_{\text{comb}}$, for methane gas.

- a. ☒ -798 kJ mol⁻¹ b. -1226 kJ mol⁻¹ c. +798 kJ mol⁻¹ d. +1226 kJ mol⁻¹

$$\begin{aligned} \Delta H_{\text{rxn}} &= \text{bonds broken} - \text{bonds formed} \\ &= (4 \cdot 415 + 2 \cdot 495) - (2 \cdot 798 + 4 \cdot 463) \\ &= -798 \text{ kJ/mol} \end{aligned}$$



10. For the boiling of methanol, CH_3OH , $\Delta H^\circ = +37.6 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = +111 \text{ J mol}^{-1} \text{ K}^{-1}$. Calculate the standard free energy change of this reaction, $\Delta G^\circ_{\text{rxn}}$.

- a. -4.5 kJ/mol b. -2.8 kJ/mol c. +2.8 kJ/mol d. ☒ +4.5 kJ/mol

$$\begin{aligned} \Delta G &= \Delta H - T \Delta S \\ &= +37.6 - 298 \cdot 0.111 \\ &= 4.5 \end{aligned}$$

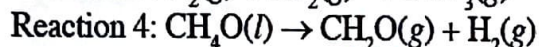
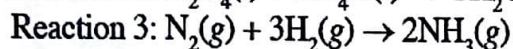
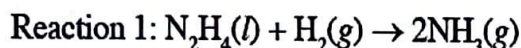
Follow up question:
Will CH_3OH boil
@ 298K?
(No... $\Delta G > 0$)



11. The reaction above is spontaneous under standard conditions, but it becomes thermodynamically unfavorable as temperature decreases. Which of the following is true?

- a. ☒ ΔS and ΔH are both negative. c. ΔS is negative, and ΔH is positive.
b. ☒ ΔS and ΔH are both positive. d. ΔS is positive, and ΔH is negative.

Use the following information to answer the next two questions.



$$\Delta H = ?$$

$$\Delta H = -37 \text{ kJ/mol}_{\text{rxn}}$$

$$\Delta H = -46 \text{ kJ/mol}_{\text{rxn}}$$

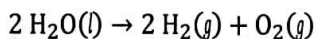
$$\Delta H = -65 \text{ kJ/mol}_{\text{rxn}}$$

12. If reaction 4 were repeated at a higher temperature, how would the reaction's value for ΔG be affected?

- ☒ a. It would become more negative because entropy is a driving force behind this reaction.
- b. It would become more positive because enthalpy is a driving force behind this reaction.
- c. It will stay the same because reaction 4 is never thermodynamically favorable.
- d. It will stay the same because reaction 4 is thermodynamically favorable at all temperatures.

13. Under what conditions would reaction 2 be thermodynamically favored?

- ☒ a. It is always favored.
- b. It is never favored.
- c. It is only favored at low temperatures.
- d. It is only favored at high temperatures.



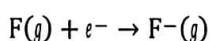
14. Based on the information in the table below, calculate $\Delta H^\circ_{\text{rxn}}$ for the reaction shown above.

Bond	Average Bond Dissociation Energy (kJ/mol)
H - H	430
O = O	500
O - H	460

- a. -900 kJ mol^{-1}
- b. -480 kJ mol^{-1}
- ☒ c. $+480 \text{ kJ mol}^{-1}$
- d. $+900 \text{ kJ mol}^{-1}$

$$\Delta H = \text{bonds broken} - \text{bonds formed}$$

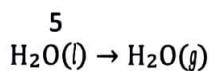
$$= (4 \cdot 460) - (2 \cdot 430 + 500)$$



FTF! Attraction!
Love wins!

15. Is the process above endothermic or exothermic? Why?

- ☒ a. Exothermic, because of the attraction between $\text{F}(g)$ and an electron.
- b. Endothermic, because of the attraction between $\text{F}(g)$ and an electron.
- c. Exothermic, because the process decreases the total number of particles from two to one.
- d. Endothermic, because the process decreases the total number of particles from two to one.



16. Which of the following is true for the above reaction?

- ☒ a. The value of ΔS is negative. ☒ c. The value of ΔH is positive.
☒ b. The value of ΔG is negative at 298 K. ☒ d. The reaction is favored at 1.0 atm and 298 K.

does water boil @ room temp? ... NO!
 (298) $\Delta G > 0$

Use the following information to answer the next two questions.

When ammonium nitrate (NH_4NO_3) dissolves in water, the temperature of the water decreases dramatically.

17. During this reaction, energy transfers from:

- a. the reactants to the products. c. the system to the surroundings.
 b. the reactants to the system. d. the products to the surroundings.

(e.) from the surroundings to the system

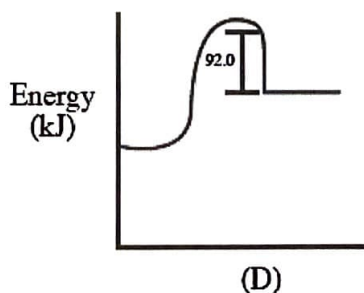
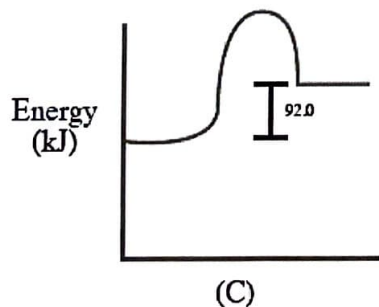
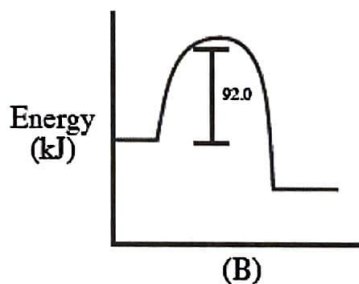
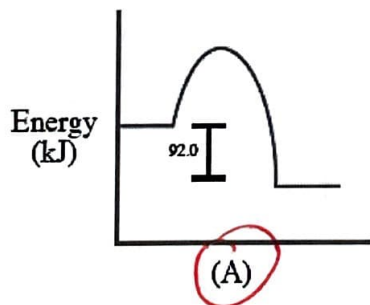
None of these are right... I made my own.

18. Which is the primary driving factor behind this reaction?

- ☒ a. Entropy $\Delta S > 0$ c. Both enthalpy and entropy
 b. Enthalpy d. Neither enthalpy and entropy



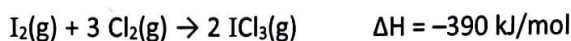
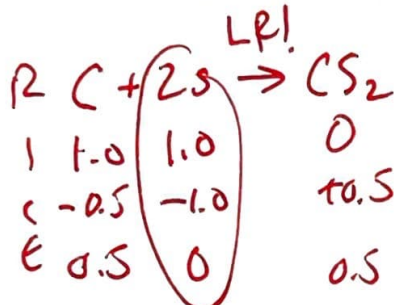
19. Which of the following energy level diagrams gives an accurate representation of the above reaction?





20. Regarding the reaction above, how much heat is absorbed or released when 1.0 mol of solid carbon reacts with 1.0 mol of solid sulfur to produce CS₂(l)? *← Reverse rxn*

- a. 46 kJ of heat is released.
- b. 46 kJ of heat is absorbed.
- c. 92 kJ of heat is released.
- d. 92 kJ of heat is absorbed.



21. Which of the following statements accurately describes the above reaction?

- a. The entropy of the products exceeds that of the reactants.
- b. I₂(g) will always be the limiting reagent.
- c. This reaction is never thermodynamically favored.
- d. The temperature of the surroundings will increase as this reaction progresses.

22. Inside a calorimeter, 100. mL of 1.0 M formic acid (HCOOH), a weak acid, and 100. mL of 2.0 M sodium hydroxide, NaOH, are mixed. The specific heat of the mixture is approximately 4.2 J/g°C, and the density is identical to that of water. If the temperature of the combined solution rises 8.0°C during the course of the reaction, calculate the approximate amount of heat released.

- a. 0.84 kJ
- b. 3.4 kJ
- c. 6.7 kJ
- d. 11.8 kJ

0.1 mol LP!!

$$q_{\text{sys}} = -q_{\text{sur}} = -m C \Delta T = -200 \cdot 4.2 \cdot 8 = -6720 \text{ J}$$

6.7

23. Consider the previous question. If the experiment is repeated with 150. mL of 1.0 M HCOOH and 50.0 mL of 2.0 M NaOH, what would happen to the values for ΔT and ΔH_{rxn}?

	ΔT	ΔH _{rxn}
(A)	Increase	Increase
(B)	Stay the same	Stay the same
(C)	Decrease	Stay the same
(D)	Stay the same	Increase

0.15 mol

0.1 mol LP!

less

Same moles will react as #22

ΔH_{rxn} does not depend on moles

24. The enthalpy change for which of the following reactions would be equal to the enthalpy of formation for ethanol (CH₃CH₂OH)?

- a. CH₃ + CH₂ + OH → CH₃CH₂OH
- b. 4 C + 6 H₂ + O₂ → 2 CH₃CH₂OH
- c. 2 C + 6 H + O → CH₃CH₂OH
- d. 2 C + 3 H₂ + ½ O₂ → CH₃CH₂OH

formation of ONE mol product